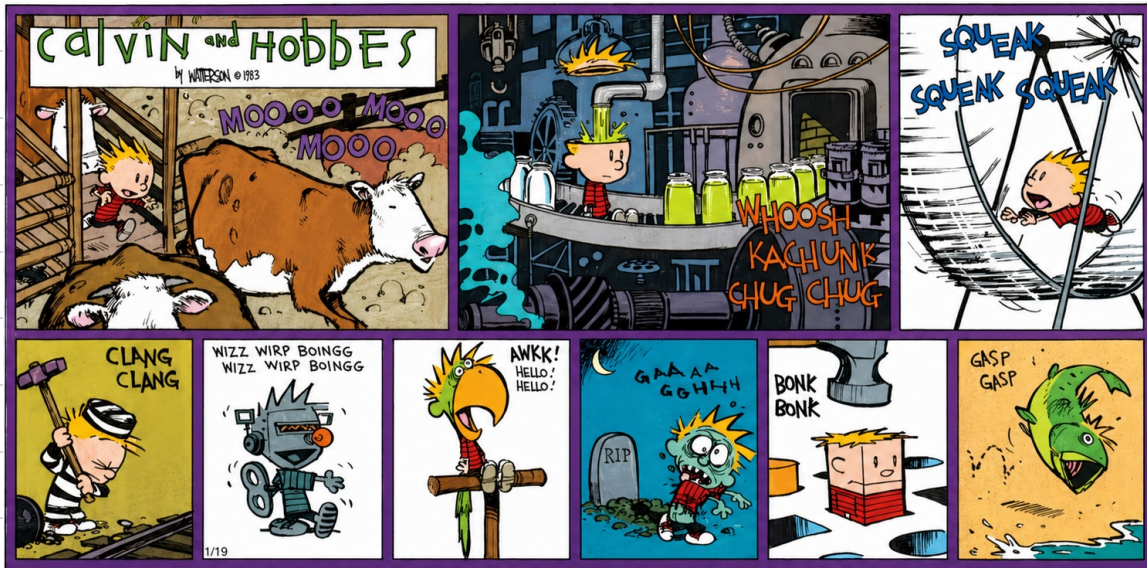
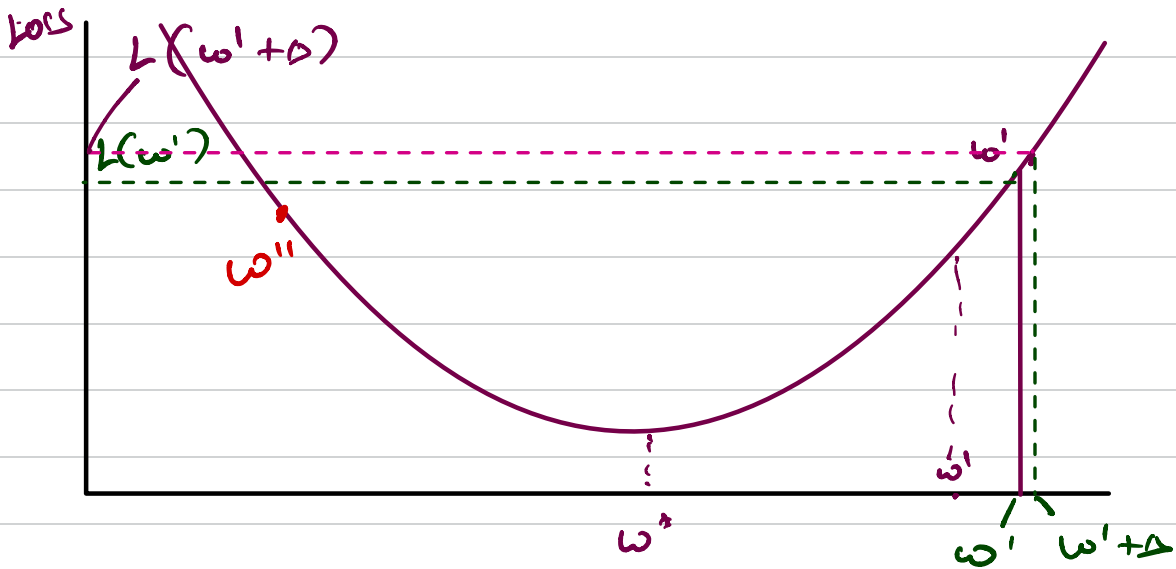






$$\text{Slope} = \frac{L(\omega' + \Delta) - L(\omega')}{\omega' + \Delta - \omega'} = \Delta \omega$$

$$\omega' = \omega' - \text{slope} \times \eta \quad \text{— step size } 0.001$$

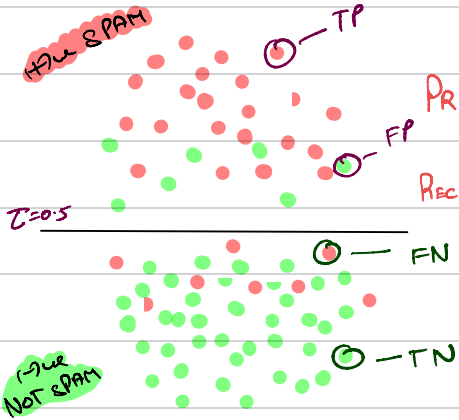
17.40 :

$$\omega_{\text{new}} = \omega_{\text{old}} - \text{slope} \times \eta$$

$$= \omega_{\text{old}} - (-\text{slope}) \times \eta$$

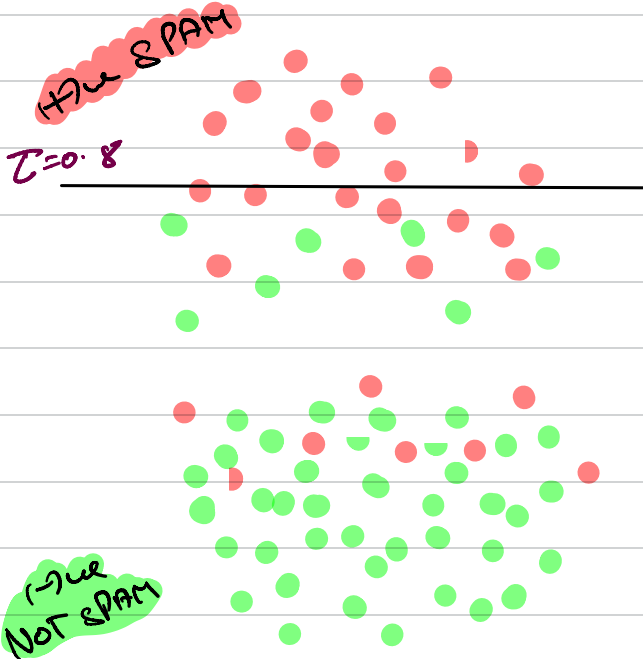
$$= \omega_{\text{old}} + \text{slope} \times \eta$$

Precision Recall - Variability



PR: $\frac{\text{What I Correctly Predicted}}{\text{Total no. of items I've predicted Positive}}$

Rec: $\frac{\text{No of classes I was able to identify}}{\text{Total Positive class elements present}}$

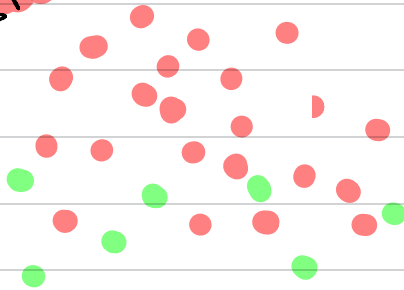


$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision = 1
Recall ↓↓

True SPAM

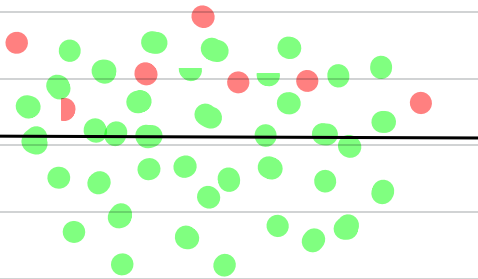


$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$\tau = 0.5$

True NOT SPAM



Recall = 1
Precision ↓↓

$$\text{Threshold} \propto \text{Precision} \propto \frac{1}{\text{Recall}}$$

F1-Score

	P	R	Avg.	Harmonic me.
M_1	0.8	0.3	0.55	0.43
M_2	0.2	0.9	0.55	0.33
M_3	0.7	0.4	0.55	0.51

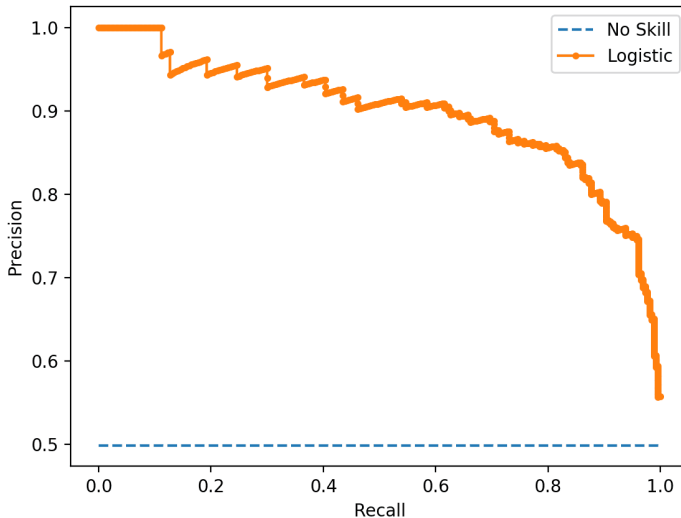
$$\text{Harmonic Mean}(a, b) = \frac{2 \times a \times b}{a + b} = \text{F Score}$$

Pos - Frow
Neg - N.Frow

FP $\rightarrow \downarrow$

FN

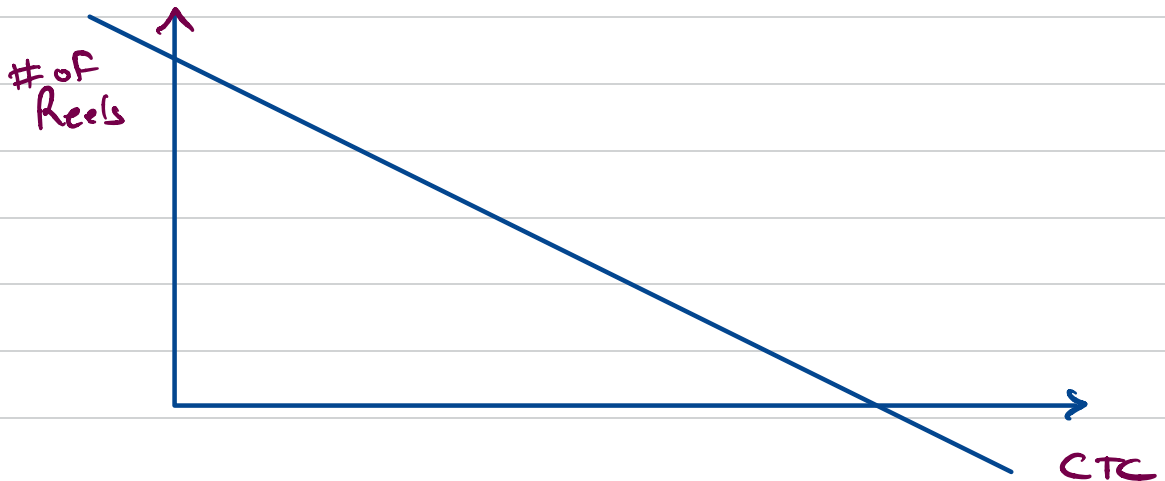
FN $\rightarrow \downarrow$



w_1 w_2 w_3 w_4 w_5 w_6 w_7 w_8 w_9

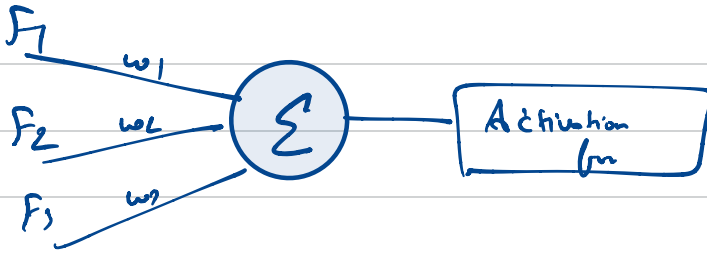
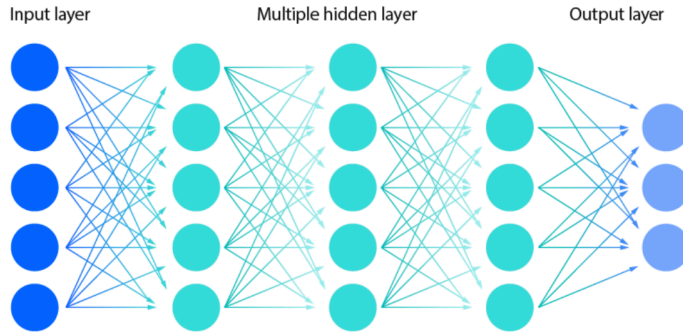
	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

$$\hat{y} = w_1 \times \text{MedInc} + w_2 \times \text{HouseAge} + \dots + w_8 \times \text{Longitude} + w_9$$



Neural Networks - Re

Deep neural network

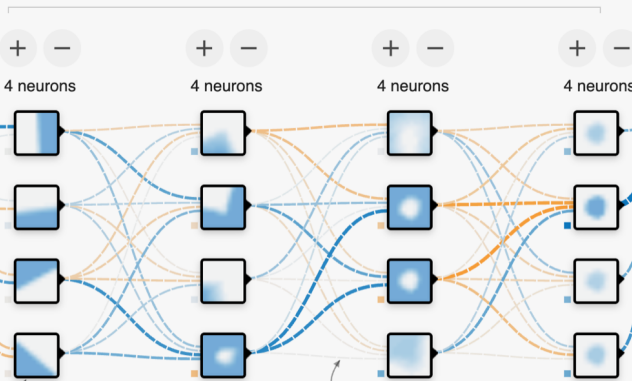


FEATURES

Which properties do you want to feed in?

- X_1
- X_2
- X_1^2
- X_2^2
- $X_1 X_2$
- $\sin(X_1)$

4 HIDDEN LAYERS

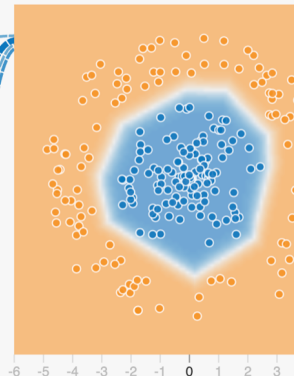


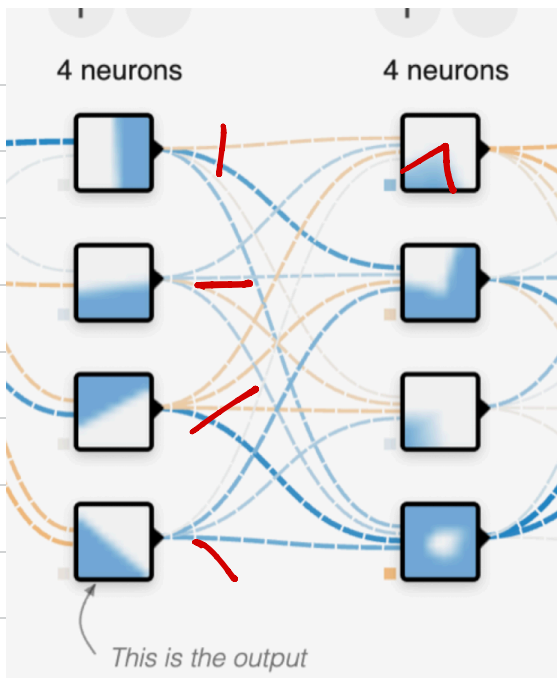
This is the output from one neuron. Hover to see it larger.

The outputs are mixed with varying weights, shown by the thickness of the lines.

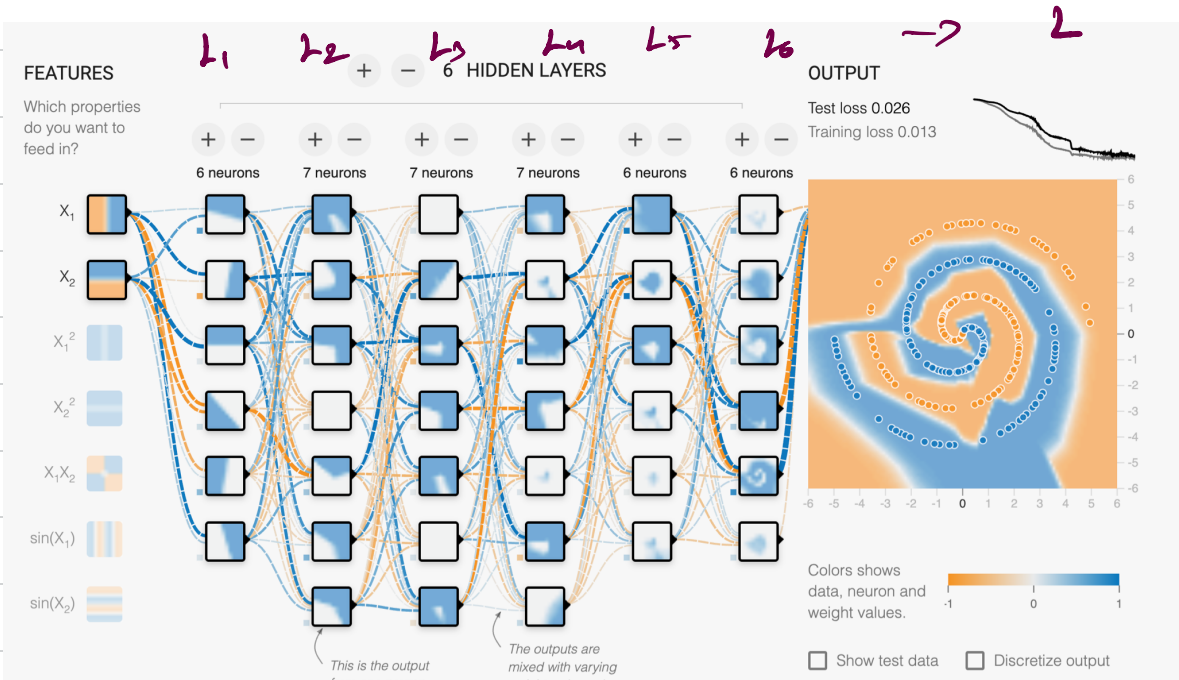
OUTPUT

Test loss 0.005
Training loss 0.003





$$\frac{\partial L}{\partial b_6} \times \frac{\partial b_6}{\partial b_5} \times \frac{\partial b_5}{\partial b_4} \frac{\partial L}{\partial b_4} \times \frac{\partial b_4}{\partial b_3} \frac{\partial L}{\partial b_3}$$



$$f(m) = \begin{cases} n & n > 0 \\ 0 & n \leq 0 \end{cases}$$

Height | Weight | Past_Genetics | HbA1c |

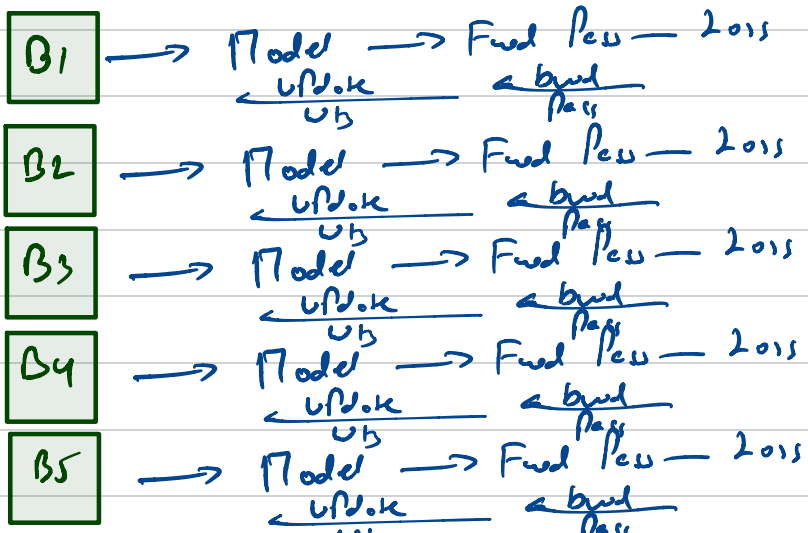
Tensorflow - Google

PyTorch - Meta/Facebook

Tensorflow - 2015 (static graph) -> Tensorflow 2.0 (2019)

PyTorch - 2-3 years late (dynamic Graph)

100 Rows of data ; batch size is 20

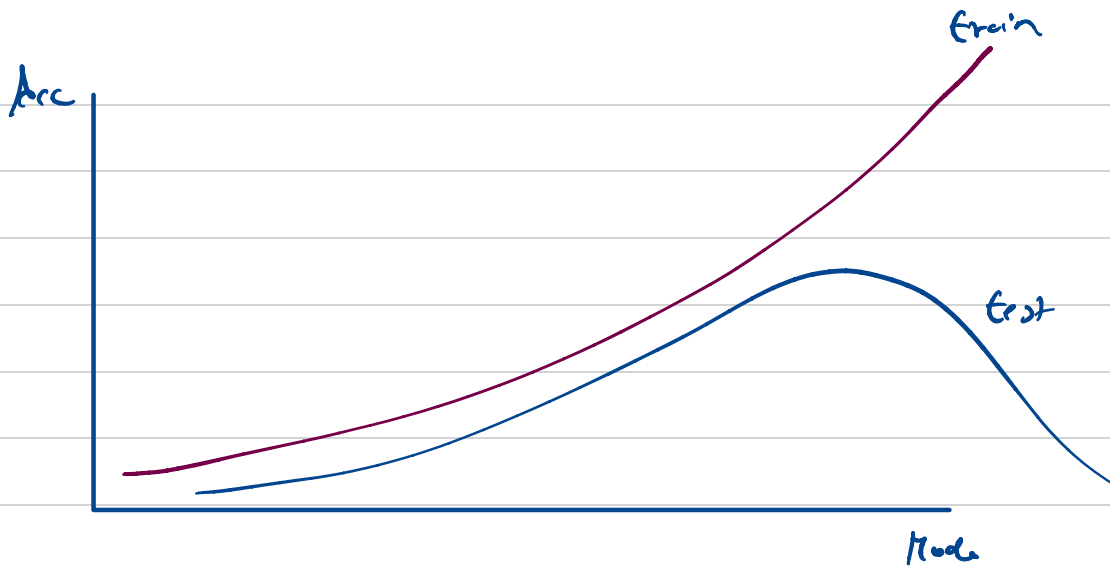


The diagram illustrates a gradient descent path on a contour plot. It features several concentric, horizontally-oriented elliptical contour lines. A red line with arrows starts from the left edge and points towards a central point marked with a cross, representing the minimum of the function.

	f_1	f_2	f_3	f_4	f_5	f_6	y
x_1							
x_2							
x_3							
x_4							
x_5							
x_6							
x_7							
x_8							

$$\begin{array}{l} O_1 \rightarrow L_1 \rightarrow G_1 \\ \rightarrow O_2 \rightarrow L_2 \rightarrow G_2 \\ \rightarrow O_3 \quad \quad \quad \cdot \quad \quad \cdot \\ O_4 \quad \quad \quad | \quad \quad | \\ O_5 \quad \quad \quad | \quad \quad | \\ O_6 \quad \quad \quad \cdot \quad \quad \cdot \\ O_7 \quad \quad \quad \cdot \quad \quad \cdot \\ O_8 \rightarrow L_8 \rightarrow G_8 \end{array}$$

$\frac{1}{8} \sum G_i$



Light bar stimulus
projected on screen

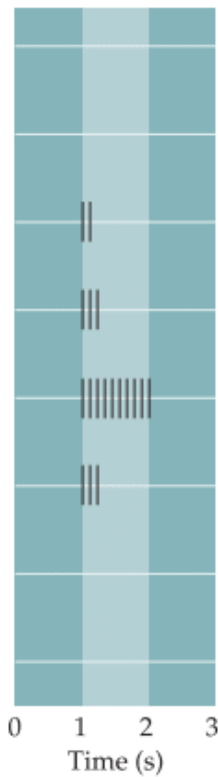


Recording from visual cortex

Record

neurons that fire together wire together

ADDICTS



$$z = wx + b$$

$$a = g(z) \quad \rightarrow \text{activation function.}$$

$$L = \frac{1}{2} (a - y)^2 \Rightarrow \frac{1}{2} \times 2(a - y) \times 1$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}$$

↓

$$(a - y) \times g'(z) \times x$$

$$\text{Relu} = (a - y) \times x \times 1 \quad ; \quad x > 0$$

$$\text{Sigmoid} = \frac{1}{1 + e^{-x}} = \frac{\sigma(x)}{1} = \frac{\sigma(x)(1 - \sigma(x))}{1}$$

↓

$$\frac{2\sigma(x)}{1}$$



$|x|$ not different

sigmoid -> tanh -> relu -> GELU

< 0

for $x < 0$

better training.

$|x|$

negative grow